

Trainer Debrief — Lab 8

Value Stream Mapping, Takt Time and the Eight Wastes — In Depth

Case study: Meridian Medical Devices — seal weld burst pressure

Objective: Build a value stream map of your capstone process and quantify waste (A2, A4).

***TRAINER ONLY.** Do not hand this out before the lab is complete.*

Expected analysis

lab-08-value-stream-map.xlsx

Available time = 2 shifts x (480 - 45) min - 30 min = 840 min = 50,400 s/day. Demand = 4200/22 = 191 units/day. TAKT = 50,400/191 = 264 s/unit. Only ONE step exceeds takt: Burst test at 360 s — it is the bottleneck and it constrains the whole line. Seal weld at 180 s is well inside takt, so the seal weld problem is a QUALITY problem, not a speed problem; do not try to fix it with more capacity. Total WIP is 1,410 units against 191/day of demand = 7.4 days of inventory sitting between steps. Value-added time is 1,235 s out of a 7.4-day lead time — a process cycle efficiency near 0.2%, which is typical and always shocking the first time a team computes it. The dominant wastes are Waiting (queues after every handoff) and Defects (the 2.9% that flows to rework), and Defects is where the capstone project attacks.

Every dataset is generated from a fixed seed, so these figures are reproducible — re-run the analysis on the shipped workbook and you will get these numbers.

What good looks like

Your VSM shows both material and information flow, you have calculated the value-added ratio and takt time from real numbers, and every logged waste carries a time or dollar quantification.

Where this sits in the capstone

A full re-derivation of value stream mapping — material AND information flow, value-added ratio, takt time and the eight wastes — then applied to your own capstone process to expose where lead time is actually consumed. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

The running process — figures that must reconcile

Checkpoint	Value	Established in
Baseline defect rate	2.9%	Lab 3
Measurement system	FAILS Gage R&R (~40% study variation)	Lab 11
Baseline capability	Pp 0.62 / Ppk 0.53	Lab 14
Proven Xs	clamp pressure, dwell time, electrode tip age	Labs 18-21
DOE optimum	clamp 47.4 bar, dwell 2.76 s	Labs 23-25
Improved capability	Ppk 1.40	Lab 27

Annual hard benefit	SGD 487,000 vs SGD 812,000 COPQ	Labs 1, 30
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If a learner's number disagrees with the column above, they have either used their own workplace data (fine — say so) or made an error (find it).

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